Printed Pages-2 University Roll No.....

End Term Examination, 2015-16
B. Tech. (CSE) II Year IV Semester
CSE4003: Computer Networks

Time: 02 ½ Hours **Total Marks: 40**

Note:
1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **1x16 = 16 MARKS**

I. Six channels, each with a 200 kHz bandwidth are to be multiplexed together. What is the minimum bandwidth requirement if each guard band is 20 kHz?

II. For n devices in a network, what is the number of cable links required for a mesh, ring, bus and star topology?

III. What is the significance of twisting in twisted pair cable?

IV. Assume a Go-Back-N protocol is used with a window size of 4 and that the ACK for packet 2 gets lost. Show the events until packet 2 is acknowledged at the sender side.

V. Consider a data stream consisting of ten Os. If this stream is encoded using the NRZ-L and Manchester encoding schemes, then what are the number of transitions that will occur in the encoded streams, respectively.

VI. What determines the size of the Ethernet frame?

VII. Is 3.54.81.76 a class A, B, C, or D IP address? What are the network ID and host ID in decimal notation? Assume that subnets are not used.

VIII. What field in the IP datagram header is used to avoid forwarding datagram's endlessly through routing loops?

IX. In the Hamming code, for a data unit of m bits how you do compute the number of redundant bits 'r' needed?

X. What is the purpose of the Address Resolution Protocol (ARP)?

XI. How can we prove that we have 2,147,483,648 addresses in class A?

XII. What is the subnetwork address if the destination address is 200.45.34.56 and the subnet mask is 255.255.240.0?

XIII. What are the three important aspects of security? Define them.

XIV. What are the fields on which the UDP checksum is calculated?

XV. What is MIME?

XVI. What are the two main categories of DNS messages?

Section B

Note: Attempt any Four Questions **3x4 = 12 MARKS**

I. Suppose we are downloading a file over 1.5 Mbps link using a TCP connection under slowstart with RTT of 100 milliseconds. What will be the window size when the link is fully utilized?

II. How the connection is established using three-way handshaking, explain in detail.

III. Name the techniques that prevent congestion. Discuss any two techniques in brief.

IV. Describe how DNS performs name resolution mentioning the role of different types of name servers.

V. Differentiate between an Active web document and dynamic web page.

Section C

Note: Attempt any Three Questions **4x3 = 12 MARKS**

I. A TCP connection uses a slowstart_threshold of 8 KB. The packet (maximum segment) size is 1 KB. Packet number 18 is lost and retransmitted. No other packet is lost. How many RTTs does it take to transfer 30 KB of data?

II. The following is the dump of a TCP header in hexadecimal format:
05320017 00000001 00000000 500207FF 00000000
   a. What is the sequence number?
   b. What is the destination port number?
   c. What is the acknowledgment number?
   d. What is the window size?

III. Explain how SMTP can handle transfer of videos and images? Also explain the advantages of IMAP 4 over POP 3 mail access protocols.

IV. Generate the public key and secret key for the following prime numbers using RSA algorithm : P = 3, Q = 11 take E = 5.

University Roll No.....

End Term Examination , Even Semester 2015-16
B.Tech.-Second Year(CSE)
Subject:-Discrete Mathematics Subject Code:-CSE-4005

Time : 2 Hours 30 Minutes **Max. Marks : 40**

Section-A

Note : Attempt all questions : **1×16=16**

I. Find whether the relation "divides" is an equivalence relation or a partial order relation on the set of integers.

II. Find the number of elements in $A_1 \cup A_2 \cup A_3$ if there are 100 elements are in A_1 , 1000 in A_2 , and 10,000 in A_3 while $A_1 \subseteq A_2$ and $A_2 \cap A_3 = \phi$.

III. Solve the recurrence relation $a_r + 3a_{r-1} = 0$.

IV. Find the sequence corresponding to the generating function $f(x) = e^{x^4}$.

V. Show that among any group of five positive integers, there are two with the same remainder when divided by 4.

VI. In how many ways, a set of two positive integers less than 100 can be chosen?

VII. Suppose that E and F are two events in a sample space and $p(E) = 1/3$, $p(F) = 1/2$, and $p(E/F) = 2/5$. Find $p(F \cap E)$.

VIII. Write the converse and contra-positive of the implication $p \Longrightarrow (\sim q \vee r)$.

IX. Write the principal DNF of $p \vee q$.

X. Show that the argument $p, p \Longrightarrow q \vdash q$ is valid.

XI. Translate the statement "some students are wise" into predicate logic .

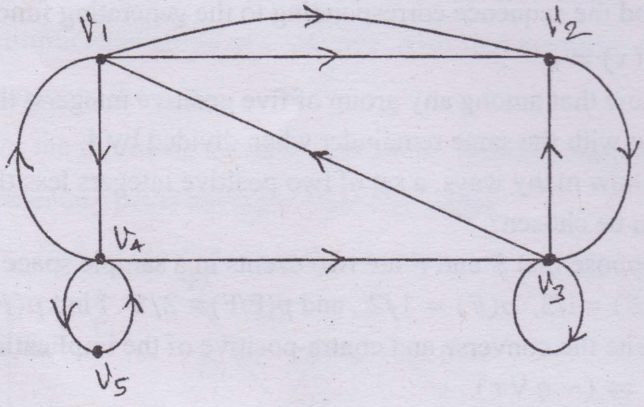
XII. Negate the statement $\forall x\, P(x) \wedge \exists y\, [Q(y) \vee R(y)]$.

XIII. Prove that intersection of any two subgroups of a group G, is a subgroup of G.

XIV. If $(R, +, .)$ is a ring then for all $a, b, c \in R$, prove that $a.(b - c) = a.b - a.c$.

XV. Draw the complete bipartite graph $K_{5,3}$.

XVI. For the following graph, verify that the sum of in-degree equals sum of out-degree equals the number of edges:



Section-B

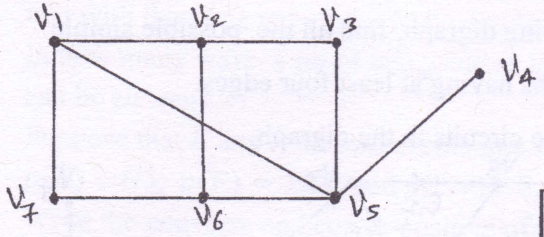
Note : Attempt any four questions : 3×4=12

I. Show that set of matrices $A_\alpha = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$ where α is a real number , forms an abelian group under matrix multiplication. Does it form a group under matrix addition ?

II. State and prove Lagrange's theorem for finite group. Hence or otherwise, show that the subset $H = \{e, a, a^2\}$ is not a subgroup of the multiplicative group $G = \{e = a^4, a, a^2, a^3\}$.

III. Let $H = \{0,2,4,6\}$ be a subgroup of the group $(G, +_8)$, $G = \{0,1,2,3,4,5,6,7\}$. Find all right cosets of H in G .Verify that any two cosets are either identical or disjoint.

IV. Show that the graph given below is a bipartite graph. Is it a complete bipartite graph? Explain.



V. Draw the graph for the adjacency matrix : $\begin{bmatrix} 0 & 3 & 2 & 2 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 2 & 2 & 1 & 0 \end{bmatrix}$

Also find the degree of each vertex.

Section-C

Note : Attempt any three questions . **4×3=12**

I. Let $f : (\mathbb{Z}, +) \to (\{1, -1\}, \times)$ is a function defined by

$$f(x) = \begin{cases} 1, & when\ x\ is\ even\ integer \\ -1, & when\ x\ ix\ odd\ integer \end{cases}$$

Show that f is a homomorphism onto. Find $kerf$.

II. Show that the set of all ordered pairs $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ of real numbers forms a ring with unity $(\mathbb{R}^2, \oplus, \otimes)$, where the binary operation addition and multiplication is defined as:

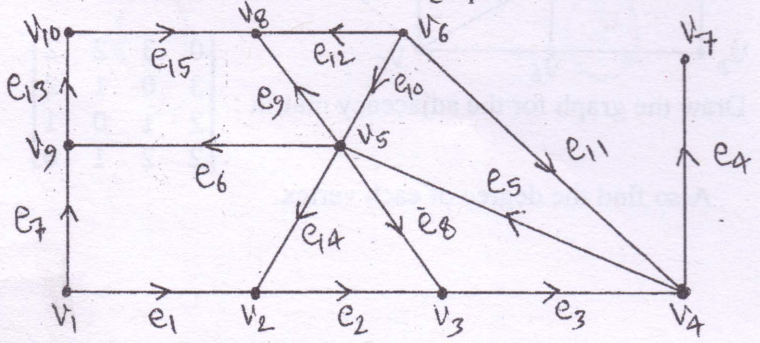
(i) $(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2), \forall\ (x_i, y_i) \in \mathbb{R}^2$

(ii) $(x_1, y_1) \otimes (x_2, y_2) = (x_1 x_2, y_1 y_2), \forall\ (x_i, y_i) \in \mathbb{R}^2$

III. Prove that the set of 5th roots of unity forms a finite multiplicative group of order five .Also find the order of each element of the group.

IV. For the following digraph, find all the possible simple elementary paths having at least four edges.

Also find all the circuits in the digraph.



SET – C

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Yr. (Even Sem.), End Term Examination, 2015-16

Subject: English for Professional Purposes
Time: 90 Minutes

Subject Code: AHE 4083
Max. Marks: 40

VERBAL APTITUDE

Spot the error in the following sentences:

1. Vacations abroad (A)/ are very inexpensive (B)/ when working (C)/ for a hotel chain. (D)/ No Error (E)
2. Having looked (A)/ through the whole music store, (B)/ the CD I wanted (C)/ just wasn't there. (D)/ No Error (E)
3. Her only (A)/ paid full time employee (B)/ is a pleasant young woman (C)/ named Rebecca. (D)/ No Error (E)
4. Cost efficient and convenient, (A)/ computer-assisted instruction is being (B)/ used by many companies (C)/ to train employees. (D)/ No Error (E)
5. Students who seek (A)/ their instructor's advice, (B)/ do often (C)/ improve their grades. (D)/ No Error (E)
6. Having submitted (A)/ the conference registration form after the deadline, (B)/ special permission from the chairperson (C)/ was needed before giving the presentation. (D)/ No Error (E)

Identify the part of sentence which contains error.

7. The government needs to provide (A)/specific regulations that create an environment where (B)/ trained scientific personnel enjoyed space and freedom (C)/ to design new products. (D) /No Error (E)
8. I cried out in agony (A)/ and throw myself on the bed (B) where I lay overwhelmed (C)/ by a rushing current of thoughts. (D)/No Error (E)
9. It is no secret that many (A)/ party workers were not happy (B)/about coalition with BJP, which has been (C)/ a bumpy ride so far for both parties. (D)/ No Error (E)
10. The terrorist attacks last year, on the Charlie Hebdo (A)/ office in January and at multiple locations in Paris in (B)/ November, have changed the way (C)/ France will look at global terrorism. (D)/ No error (E)
11. With nominal GDP growing at (A)/ a lower rate than the real GDP, further addition (B)/ to debt increase debt-GDP (C)/ ratio at a much faster rate. (D)/ No error (E)
12. This is only a beginning and does not (A)/ negates the possibility of (B)/ another kind of experiment (C)/ emerging in the higher echelons of art. (D)/ No Error (E)

Identify the part of sentence which contains the error.

13. Much emphasis has placed (A)/on the benefits of (B)/ uniformity that the Goods and Services Tax Bill (C)/ would usher in. (D)/ No Error (E)
14. Almost five years after legislation to regulate contracts (A)/ between builders and buyers was moot, (B)/ it is expected to be endorsed by Lok Sabha this (C)/ week after it was cleared by the Upper House last week. (D)/ No Error (E)
15. The strident opposition to EPF tax (A)/ must read in the context of (B)/ the virtual absence of a social security net (C)/ of any worth in India. (D)/ No Error (E)
16. The higher judiciary should also identify (A)/ the root causes of acquittals and (B)/ ways in which the child-friendliness of (C)/ special courts can be enhance (D)/ No error (E)
17. A committee constituted by the Central Board of Direct Taxes (A)/ to examine the business models for e-commerce, (B)/ identify the direct tax issues in relation to (C)/ e-commerce transactions and provides suggestions. (D)/ No Error (E)
18. Renowned scientist K Vijayraghavan has given (A)/ additional charge of Secretary, Department of Scientific and Industrial Research (DSIR) (B)/ for three months as (C)/ the post has been lying vacant for over one year. (D)/ No Error (E)

Decide which sentence is the most preferable with respect to grammar/ meaning and usage, for formal writing.

19. A. We try to let go of our thoughts about work hence they creep into our conversation.
    B. We try to let go of our thoughts about work so that they creep into our conversation.
    C. We try to let go of our thoughts about work as a result they creep into our conversation.
    D. We try to let go of our thoughts about work yet they creep into our conversation.
    E. None of the above is correct.
20. A. As we count our steps and watch the time, we pass by sculptures and artwork along the walkway.
    B. After we count our steps and watch the time, we pass by sculptures and artwork along the walkway.
    C. Before we count our steps and watch the time, we pass by sculptures and artwork along the walkway.
    D. Likewise, we count our steps and watch the time, we pass by sculptures and artwork along the walkway.
    E. None of the above is correct.
21. A. Sitting back down in our offices, we are bothered by both co-workers or bosses; our minds are still filled with views of the bay and the smell of the sea air.
    B. Sitting back down in our offices, we are bothered by none co-worker; our minds are still filled with views of the bay and the smell of the sea air.
    C. Sitting back down in our offices, we are bothered by just as co-workers or bosses; our minds are still filled with views of the bay and the smell of the sea air.
    D. Sitting back down in our offices, we are bothered by neither co-workers nor bosses; our minds are still filled with views of the bay and the smell of the sea air.
    E. None of the above is correct.
22. A. When they entered the house, they found that it had been ransacked. Even though nothing was missing, they called the police and reported that their house had been broken into.
    B. When they entered the house, they found that it had been ransacked. But nothing was missing,

they called the police and reported that their house had been broken into.

C. When they entered the house, they found that it had been ransacked. While nothing was missing, they called the police and reported that their house had been broken into.

D. While they entered the house, they found that it had been ransacked. As nothing was missing, they called the police and reported that their house had been broken into.

E. None of the above is correct.

23. A. When all of them had gone out, he was alone in the house. He locked the door as he would feel safer, and switched on the television set.

B. When all of them had gone out, he was alone in the house. He locked the door so that he would feel safer, and switched on the television set.

C. When all of them had gone out, he was alone in the house. He locked the door because he would feel safer, and switched on the television set.

D. When all of them had gone out, he was alone in the house. He locked the door so that he would feel safer, but switched on the television set.

E. None of the above is correct.

24. A. You might not leave small objects lying around. Such objects might be swallowed by children.

B. You could not leave small objects lying around. Such objects may be swallowed by children.

C. You would not leave small objects lying around. Such objects will be swallowed by children.

D. You should not leave small objects lying around. Such objects may be swallowed by children.

E. None of the above is correct.

25. A. If he hadn't told the truth, we would have found our eventually.

B. If he hadn't been told the truth, we would have found our eventually.

C. If he hadn't told the truth, we would found our eventually.

D. If he hadn't told the truth, we would be found our eventually.

E. None of the above is correct.

26. A. Unless my watch is wrong, we would arrive at exactly 2 pm.

B. Unless my watch was wrong, we will arrive at exactly 2 pm.

C. Unless my watch is wrong, we will arrive at exactly 2 pm.

D. Unless my watch was wrong, we would arrived at exactly 2 pm.

E. None of the above is correct.

27. A. If I have had more money, I would have bought more food.

B. If I had more money, I would have bought more food.

C. If I had had more money, I would bought more food.

D. If I had had more money, I would have bought more food.

E. None of the above is correct.

28. A. If the storm hit Mathura, thousands of people might have lost their lives.

B. If the storm had hit Mathura, thousands of people might have lost their lives.

C. If the storm had hit Mathura, thousands of people might lost their lives.

D. If the storm had been hit Mathura, thousands of people might have lost their lives.

E. None of the above is correct.

29. A. I have no fire insurance. If my house burnt down, I would have no way to buy new furniture.

B. I have no fire insurance. If my house was burnt down, I would have no way to buy new furniture.

C. I have no fire insurance. If my house was burnt down, I would have been no way to buy new furniture.

D. I have no fire insurance. If my house will burnt down, I would have no way to buy new furniture.

E. None of the above is correct.

30. A. If you will pay heed to the small details, the general plans will surely succeed.

B. If you pay heed to the small details, the general plans will surely succeed.

C. If you would heed the small details the general plans would have succeeded.

D. If you heed to the small details the general plans would surely succeed.

E. None of the above is correct.

31. Identify the sentence in which all the underlined words have been correctly tagged.

A. The snatchers pocket (noun) the gold-chains (noun) and disappear (verb) in the crowd.

B. They were holidaying (noun) in France when they heard that there was a coup in their country.

C. For that price (noun) it is a good (adjective) buy (verb).

D. She backed (verb) the car (noun) into the parking lot (noun).

E. The old woman willed (verb) her property (adjective) to an orphanage (noun).

32. Identify the incorrect part of speech tagged-

A. Bedevil (noun)
B. Ennoble (verb)
C. Wreckage (noun)
D. Bookish (adj)
E. Trickery (noun)

33. Which of the following words is not derived/ inflected out of the root word "**Trans**"?

A. Trance
B. Transference
C. Transceiver
D. Transcription
E. None of these

34. Which one is not an example of **inflection**?

A. Superego (ego)
B. Surplus (plus)
C. Intercollegiate (collegiate)
D. Proliferation (proliferate)
E. None of these

35. Identify the correct part of speech of the emboldened word in the sentence- "First on my list, but not necessarily **first** in importance, I would mention intellect."

A. Adjective
B. Adverb
C. Verb
D. Noun
E. None of these

36. Identify the category of the compound words – **"Tightrope"**

A. Adjective
B. Verb
C. Noun
D. Preposition
E. Interjection

Provide the most suitable choice for the blank spaces in Q. No. 37-42.

37. The authorities studied the problems of the students at GLAU with a view to _____ academic environment of the campus.

A. Improve
B. To improve
C. Improving
D. Have improved
E. None of the four

38. _____ a bath, Raghunath went to his college.

A. Having taken
B. Taking
C. Having
D. Bathing
E. None of the four

39. The railway is increasing fares with a view to _____ better facilities to the passengers,
   A. Providing
   B. Provide
   C. Have provided
   D. Provided
   E. None of the four

40. I am so tired and I feel like _____.
   A. Sleeping
   B. Sleep
   C. To sleep
   D. Slept
   E. None of the four

41. Reading newspaper will help you _____ for the competitive tests.
   A. Prepare
   B. Preparing
   C. To prepare
   D. Preparation
   E. None of the four

42. When I was returning from my college, I saw Ravi _____ towards Buddha Park.
   A. Go
   B. Going
   C. To go
   D. Had gone
   E. None of the four

Choose the correct 'causative' option from the choices (Q. No. 43-48).

43. A. The businessman asked his secretary make the copies of the report.
   B. The businessman had his secretary make the copies of the report.
   C. The businessman had to make the copies of his secretary's report.
   D. The businessman had his secretary to make the copies of the report.
   E. The businessman got the copies of the report made by his secretary.

44. A. She always gets my help with her homework.
   B. She always gets to me to help with her homework.
   C. She always get me to help with her homework.
   D. She always gets me to help with her homework.
   E. None of the above.

45. A. I am going to a friend of mine, who is an electrician to repair my DVD player.
   B. I am going to have my DVD player repaired next week by a friend of mine, who is an electrician.
   C. I am going to repair my DVD player next week by a friend of mine, who is an electrician.
   D. A friend of mine, who is an electrician, is going to repair my DVD player next week.
   E. None of the above.

46. A. Ralph has made us promise not to tell what he did before he entered the forbidden zone.
   B. Ralph had made us promise not to tell what he did before he entered the forbidden zone.
   C. Ralph had made promise to us not to tell what he did before he entered the forbidden zone.
   D. Ralph had made promise not to tell what he did before he entered the forbidden zone.
   E. None of the above.

47. A. The Johnsons will always make us feed their dog for them during their summer vacation.
   B. The Johnsons will always have us fed their dog for them during their summer vacation.
   C. The Johnsons will always have had us feed their dog for them during their summer vacation.
   D. The Johnsons would have always let us feed their dog for them during their summer vacation.
   E. None of the above.

48. A. I could get to my brother to feed the cows every evening.
   B. I make my brother feed the cows every evening.
   C. I can get my brother fed the cows every evening.
   D. I have taken my brother to feed the cows every evening.
   E. None of the above.

Keeping in view the agreement of the verb with the subject, find out the only correct choice (Q. No. 49-54).

49. A. Each of these women wishes that she had read the care instructions before washing the delicate and expensive dresses.
   B. Each of these women wish that she had read the care instructions before washing the delicate and expensive dresses.
   C. Each of these women has wish that she had read the care instructions before washing the delicate and expensive dresses.
   D. Each of these women wishing that she had read the care instructions before washing the delicate and expensive dresses.
   E. Each of these women are wishes that she had read the care instructions before washing the delicate and expensive dresses.

50. A. Here is the mail and newspapers that I picked up for you while you were on vacation.
   B. Here are the mail and newspapers that I picked up for you while you were on vacation.
   C. Here be the mail and newspapers that I picked up for you while you were on vacation.
   D. Here has the mail and newspapers that I picked up for you while you were on vacation.
   E. Here have the mail and newspapers that I picked up for you while you were on vacation.

51. A. Mr. Khosla, our English teacher, believes that the students who major in economics or physics ruins their imaginations.
   B. Mr. Khosla, our English teacher, believes that the students who major in economics or physics ruin their imaginations.
   C. Mr. Khosla, our English teacher, believes that the students who major in economics or physics ruining their imaginations.
   D. Mr. Khosla, our English teacher, believes that the students who major in economics or physics has ruin their imaginations.
   E. Mr. Khosla, our English teacher, believes that the students who major in economics or physics have ruins their imaginations.

52. A. Neither of Freud's parents has much intelligence when it comes to choosing spouses. Freud's father has married five times, and Mom just presented him with stepfather number three.
   B. Neither of Freud's parents have much intelligence when it comes to choosing spouses. Freud's father has married five times, and Mom just presented him with stepfather number three.
   C. Neither of Freud's parents are showing much intelligence when it comes to choosing spouses. Freud's father has married five times, and Mom just presented him with stepfather number three.
   D. Neither of Freud's parents are much intelligence when it comes to choosing spouses. Freud's father has married five times, and Mom just presented him with stepfather number three.
   E. Neither of Freud's parents is much intelligence when it comes to choosing spouses. Freud's father has married five times, and Mom just presented him with stepfather number three.

53. A. Here stands Tonya and Robert, the two students who scored 100 percent on the subject-verb agreement quiz.

B. Here stand Tonya and Robert, the two students who scored 100 percent on the subject-verb agreement quiz.
C. Here is standing Tonya and Robert, the two students who scored 100 percent on the subject-verb agreement quiz.
D. Here has stands Tonya and Robert, the two students who scored 100 percent on the subject-verb agreement quiz.
E. Here have stands Tonya and Robert, the two students who scored 100 percent on the subject-verb agreement quiz.

54. A. Who needs a pet if you live in Florida? There already is lizards that sneak in between the window screens and cockroaches that take up residence in every bathroom and kitchen.
B. Who needs a pet if you live in Florida? There already are lizards that sneak in between the window screens and cockroaches that take up residence in every bathroom and kitchen.
C. Who needs a pet if you live in Florida? There already has lizards that sneak in between the window screens and cockroaches that take up residence in every bathroom and kitchen.
D. Who needs a pet if you live in Florida? There already was lizards that sneak in between the window screens and cockroaches that take up residence in every bathroom and kitchen.
E. Who needs a pet if you live in Florida? There already has been lizards that sneak in between the window screens and cockroaches that take up residence in every bathroom and kitchen.

Find out one word substitute for expressions given in Q. No. 55-60.

55. Place where people grow fruit trees
A. Orbit D. Orchard
B. Ordeal E. None of the four
C. Otiose

56. Defeat (someone) easily and completely
A. Rue D. Rustle
B. Rout E. None of the four
C. Rusticate

57. Business of making fashionable and expensive clothes for women
A. Haute couture D. Coterie
B. Debonair E. None of the four
C. Boutique

58. View or sight that looks like a picture
A. Tabloid D. Tableau
B. Graffiti E. None of the four
C. Dekko

59. Plan of a journey, including the route and the places that you visit
A. Folio D. Topography
B. Ploy E. None of the four
C. Itinerary

60. Period of temporary economic decline during which trade and industrial activity are reduced
A. Inception D. Tete-e-tete
D. Locus standi E. None of the four
B. Recession

Read the passage carefully and answer the questions given below (Q. No. 61-65).

Of populism and prohibition

Across India, States are marching towards total prohibition. Kerala has embarked on a 10-year path to total prohibition, by first limiting the sale of "hard liquor" to five-star bars and restaurants and gradually reducing the number of sale points. Bihar initiated a two-step plan, with a ban on country liquor effective from April 1, followed just days later with a prohibition on the sale of Indian Made Foreign Liquor, bureaucratese for everything other than country brew. Indeed, in a sign of the moral panic that grips political parties at the mention of prohibition, the Bihar Assembly's successful adoption of the Bill banning country liquor was accompanied by a unanimous resolution by MLAs that they would not consume alcohol. Now a similar competitiveness is playing out in the Tamil Nadu Assembly election campaign. The ruling AIADMK has countered the DMK's pledge to turn the State dry by affirming a graded shift. This is not the first time India has grappled with the social consequences of alcohol consumption, such as alcoholism, indebtedness and domestic violence. For example, vestiges of prohibition-era practices survive in States such as Maharashtra, though the law tends to realistically look the other way; Gujarat of course continues to remain dry. In recent decades, States such as Haryana and undivided Andhra Pradesh adopted a prohibitory regime, but abandoned it soon after. Whether the current spate of prohibition legislation will sustain is unclear. But once again, the populist solution of prohibition is being offered without attendant focus on the social problems that it seeks to address.

The creep of the nanny state to guard citizens from their worst selves, or at least their lack of self-discipline, is worrying. It is perhaps the overhang of the Gandhian spirit of the Directive Principles of State Policy in the Constitution that inhibits politicians and civil society from shedding hypocrisy and initiating public advocacy of moderation. However, there is a pragmatic case against prohibition as well. Banning the sale and consumption of alcohol has, in this country's experience, not been an effective check against its use. It has only criminalised the activity, with disastrous consequences for individual health, the economy and administration — these include bootlegging, liquor mafias, spurious liquor, and a complicit police. It also deprives States of an important source of revenue. For instance, in Tamil Nadu nearly Rs.30,000 crore, or over a quarter of its revenue in 2015-16, came from taxes on the sale of alcohol and excise on manufacturing spirits. This income has enabled successive regimes from 2006 onwards to splurge on social sector schemes, especially the trademark programmes to supply free rice to nearly all ration card holders, distribute consumer goods and maintain its pioneering nutritious noon meal scheme for all children in government and aided schools and anganwadis. Certainly, alternative sources of revenue must be found if prohibition can virtuously, magically transform society. That case has, however, not been made, in argument or by experience. **(The Hindu – April 20, 2016)**

61. According to the author, the politicians, as well as the society should –
A. Uphold the Gandhian spirit of the Directive Principles of State Policy.
B. Initiate public advocacy of moderation.
C. Ban the sale and consumption of alcohol.
D. Invest the earned revenue in strengthening the country's security measures.
E. Both 'A' and 'D'.

62. The author is of the opinion that –
A. Prohibition, at present, cannot transform our society magically.
B. The government is losing more than one-fourth of its revenue.
C. Prohibition is engendering illegal liquor markets.
D. The government and the leaders are using prohibition to create a political sensation.
E. All of the above.

63. "The moral panic," that the author talks about, gets validated through –

A. Kerala's 10-year path to total prohibition.
B. Gujarat's ban on liquor.
C. The resolution of the MLAs to stop consumption of alcohol.
D. Bihar Assembly's successful adoption of the Bill on the prohibition of alcohol.
E. Both 'C' and 'D'.

64. Which of the following sentences correctly use the word "complicit" in the same sense as in the article?
A. He will never share his complicit with anyone else.
B. Her complicit gets revealed very naturally.
C. The complicit penguin dived into the water.
D. Doping involves people complicit in the effort.
E. None of the above.

65. In this article, the word "vestiges" refers to –
A. Traces.
B. Failures.
C. Achievements.
D. Redundancies.
E. Possibilities.

Read the passage carefully and answer the questions given below (Q. No. 66-70).

Florida's "Everglades", the largest national park in the eastern USA, are in serious danger. One of the most significant areas of "wetland" in the world, and classified by UNESCO as a "World Heritage Site", the park has suffered dramatically over the last sixty years on account of encroaching agriculture and the vast expansion of Florida's population. If the decline is not stopped, the entire ecosystem of southern Florida could be irreparably damaged – and the sunshine state, one of the fastest growing states in the USA, could become an inhospitable wasteland.

Back in the year 2000, in order to prevent just such a catastrophe, and ensure the survival of America's most distinctive National Park, Congress approved a 30-year 8.2 billion dollar Everglades rescue plan, the largest concerted environmental project the world had ever known.

As a "wetland" region, the whole environment of the Everglades depends on the flows of water into and out of its vast area of natural swampland. Before modern agriculture and drainage schemes began to change the hydrology of the whole of southern Florida, water that fell on the central part of the Florida Peninsula used to flow regularly and naturally into the great Lake Okeechobee, the large lake at the northern end of the Everglades region. From there, it would flow out slowly and regularly in a southerly direction, through the Everglades, and into the Gulf of Mexico.

Since the 1940's, this natural system has been radically modified. Over 1,700 miles of canals have been built, in order to control the risk of flooding, to provide water for Florida's growing population, and drain land for agriculture and house-building. Today a large amount of the water that flows into Lake Okeechobee is rapidly evacuated through man-made canals, out into the Atlantic Ocean.

The Everglades rescue program was prompted by the publication in 1998 of a 4000 page report, produced by the U.S. Army Corps of Engineers. According to the report, the Central and Southern Florida (C&SF) water supply system was designed to serve 2 million people, but more than 6 million people now live in the area, and the population is expected to reach 12 to 15 million by 2050. This explosive growth has strained the system's ability to perform its intended functions, and the Everglades are seriously threatened.

66. The passage aims to:
A. Generate a consciousness among the people of America the need to save ground water.
B. Make the people nonchalant of the Everglades rescue plan.
C. Show that Americans are concerned regarding their environment.
D. Analyze the impact of human development and its impact on Nature.
E. Both 'B' and 'D'.

67. The word "concerted" in the article does NOT mean –
A. Visited
B. United
C. Individual
D. Synergetic
E. Both 'A' and 'C'

68. Which of the words given below is antonymous to "radically"?
A. Dolefully
B. Woefully
C. Mournfully
D. Remorsefully
E. None of the above

69. What does the word "swampland" in the passage mean?
A. Occupied land
B. Dry land
C. Irrigated land
D. barren land
E. None of the above

70. The word "strained" in the passage means –
A. Inundated
B. Exhausted
C. relegated
D. nauseated
E. Both 'A' and 'C'

Spot the error in the following sentences (Q. No. 71-80).

71. The sudden noise (A)/ frightened the baby (B)/ and made it (C)/ to cry. (D)/ No Error (E)
72. The dog wagged (A)/ it's tail (B)/when it saw me (C)/ approaching. (D)/ No Error (E)
73. There was signs (A)/ of neglect (B)/ showing that (C)/ no one was living there. (D)/ No Error (E)
74. He doesn't want (A)/ to get any credit (B)/ with whatever good (C)/ he does to others (D)/ No Error (E)
75. We have observed that (A)/ many good programmes (B)/ suffer of shortage (C)/ of funds and other resources. (D)/ No Error (E)
76. Mahesh and Rahul live (A)/ next by the post office (B)/ not far (C)/ away from the station. (D)/ No Error (E)
77. Government should use (A)/the occasion well and (B)/extend an olive (C)/branch for opposition parties. (D)/ No Error (E)
78. You deserve to be (A)/ rewarded by your successful (B)/ completion of the entire work (C)/ in such a short time (D)/ No Error (E)
79. He never attended lectures (A)/ and used to borrow (B)/ the books of his friends (C)/ to study for the examination. (D)/ No Error (E)
80. Our market research shows (A)/ that there will be (B)/ a huge demand by such (C)/ products in the next few years. (D)/ No Error (E)

End Term Examination,
Even Semester 2015-16

Course: B. Tech. II Yr. **Univ. No.** ----------
Title of Paper: Environmental Studies
Paper Code: AHS-3001
Time: 2.5 Hours **Max. Marks**: 40

Section-A

Note: Attempt All Questions **1 X 16 = 16 Marks**

(I) What is ecological succession?
(II) What is ecological geothermal energy?
(III) Define boundary layer.
(IV) Which layer of atmosphere has umbrella layer?
(V) What do you mean by in-situ conservation?
(VI) What is biosphere reserve?
(VII) Define intangible resources with examples.
(VIII) Write two chemicals used for de-fluorination of water.
(IX) What is sanitary land-filling?
(X) Define population explosion.
(XI) What is albedo?
(XII) What are basic theme of Montreal protocol?
(XIII) Write extended form of WWF and BNHS.
(XIV) Write value of ecological foot print for US, China and India.
(XV) When was water (prevention and control of pollution) act enacted in India?
(XVI) Write two water borne diseases caused by bacteria.

Section-B

Note: Attempt Any four Questions **3 x 4 = 12 Marks**

(I) Write roles of NGO in protection of environment.
(II) Discuss levels and dimensions of environmental education.
(III) Write a note on life cycle analysis (LCA).

(IV) What are the guidelines for excellence in environmental education?

(V) Why is environmental education provided to engineers?

Section-C

Note: Attempt Any Three Questions **4 x 3 = 12 Marks**

(I) Discuss the power of central government under environmental protection act 1986.

(II) Discuss global efforts in environmental protection?

(III) What are the principles of EIA?

(IV) What is environmental ethics? Give important eco-centric points for harmonic relation of human with nature.

End Term Examination, 2015-16
Course: B. Tech. (CSE), Year: II, Semester: IV
Introduction to Microprocessor (CSE 4002)

Uni. Roll No:.....

Time: 150 Min. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. What is the function of microprocessor in a system?
II. Differentiate Processor Cycle and Machine Cycle.
III. What is the name of machine cycle if IO/M =1, S1=0 and S0=1?
IV. What is the difference between microprocessor and microcontroller?
V. Which signal causes the microprocessor to immediately terminate its present activity?
VI. What is the addressing mode used in instruction MOV M, C?
VII. If the operating frequency is 5MHz then time required for 1 T-state is.....
VIII. Name the different types of flags used in 8085 microprocessor.
IX. What is interrupt service routine?
X. What do you mean by Non-Maskable interrupts?
XI. What is the position of the Stack Pointer after the POP instruction?
XII. Why interfacing is needed for 1/0 devices?
XIII. What is the control word in BSR mode to set PC4?
XIV. All the functions of the ports of 8255 are achieved by programming the bits of an internal register called as ....
XV. In cascaded mode, the number of vectored interrupts provided by 8259A is.....
XVI. Which microprocessor pins are used to request and acknowledge a DMA transfer?

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Name the modes used by the DMA processor to transfer data and explain each transfer mechanism in short.

II. List the applications of programmable interval timer.

III. What is the purpose of control word written to control register in 8255?

IV. Explain in detail the operating modes of PPI? Also explain Interfacing.

V. Differentiate memory mapped I/O and I/O mapped I/O.

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. Draw and explain the block diagram of 8254 programmable interval timer. Also explain the various modes of operation.

II. Discuss and explain interfacing of 8 bit input port with any another chip or processor in detail.

III. Draw and discuss the block diagram of the internal architecture of PIC 8259.

IV. What is the use of 8237 IC? Discuss various transfer modes of 8237.

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End-Term Examination, 2015-16
Course: B.Tech, Year: II Semester: IV
Object Oriented Programming (CSE 4004)

Time: 2.5 Hours **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

(I) What are different UML relationships?
(II) Explain the features of abstract class?
(III) Explain Applet life cycle methods?
(IV) Enlist different components used in case of Event Handling for Button?
(V) Differentiate between keywords final and static?
(VI) Design a java Applet program to print " Java Applet" in Red color with Font Times New Roman size 18 background color as Green?
(VII) Explain the use of keyword super with an example?
(VIII) What is Main Thread in java? Explain two reasons why Main thread is important?
(IX) Design a java program to print the default name of main thread along with default property and to change the name of main thread to name "My Main Thread"?
(X) Explain the concept of monitor in java? Name the method used for interprocess communication mechanism in Threading?

(XI) Enlist any six attributes used in case of <applet> tag? Name the method used to produce output message to the status window of the browser or applet viewer?

(XII) Name three methods available in KeyListener Interface? Enlist the two methods each for MouseListener and MouseMotionListener Interface?

(XIII) Explain the concept of Bean Introspection along with any one methods of Introspector class?

(XIV) Discuss the use of java servlets? Enlist the ways of creating servlets?

(XV) Discuss the significance of Stub & Skeleton in RMI? Give syntax with description to create JAR file?

(XVI) Write the steps for creating Java Bean? Enlist any two properties of java Bean?

Section- B

Note: Attempt any Four Questions **(3x4=12)**

I. Explain the term Enterprise java Beans and its use? Discuss its types with any example?

II. Explain RMI layered architecture? Discuss the role of RMI registry in creating RMI applications?

III. Design a JDBC program to insert values in a table student having fields "Name , Age", "Roll_No" and select the inserted values?

IV. Explain servlet life cycle? Discuss the task performed by java servlet?

V. Differentiate between:
a) Statement & Prepared Statement of JDBC
b) accessor() method and mutator() method of Java Bean

Section- C

Note: Attempt any Three Questions **(4x3=12)**

I. Explain Driver Manager class and its usages in JDBC? Write the steps along with syntax to create JDBC application. Discuss the benefits of Type III Driver and use of callable statement in JDBC?

II. Write six differences between Generic Servlet & HTTP Servlet? Discuss briefly with syntax the suitable technique for creating servlet? Name any two server used in case of Servlet programming?

III. Design a RMI program to compute square root of any number? Enlist the steps of the program specifying the use of different predefined classes and their methods used in the program?

IV. Discuss the advantages of distributed applications? Explain the transport layer's responsibility in RMI? Enlist the advantages of Type IV driver?

END Term Examination, 2015-16
Course: B.Tech. (CSE), Year: II, Semester: IV

Operating System (CSE 4001)

Uni. Roll No:.....

Time: 2.5 Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. List the four steps that are necessary to run a program on a completely dedicated machine.

II. Write any two differences between trap and an interrupt.

III. What are the main advantages of the microkernel approach to system design?

IV. Differentiate Multiprogramming and Multitasking. (Any two)

V. What are the two conditions under which a parent process may terminate the execution of one of its children process?

VI. Explain the significance of PCB when CPU switches from one user to another.

VII. What are the fundamental requirements to be met by concurrent processes in order to cooperate correctly using the shared data?

VIII. What is race condition?

IX. Explain binary semaphore.

X. Define livelock.

XI. What is RAG?

XII. Define the need of Mutual Exclusion.

XIII. What do you mean by load time binding?

XIV. How can we reduce the memory overhead of a paged system?

XV. Why is page size always a power of 2?

XVI. What is thrashing?

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Under what circumstances do page fault occurs? Describe the actions taken by the operating system when a page fault occurs.

II. What hardware support are needed to implement virtual memory? Given a 4GB of virtual space and typical page size of 4KB. How many virtual pages would this imply? If each page table entry is 5 bytes, how much space is required for the whole page table?

III. Describe various file allocation strategies. Also discuss the criteria that have to decide which strategy is to be adopted for a particular file.

IV. Compare Paging and Segmentation. For a paged system, TLB hit ratio is 0.9. let the RAM access time, t be 20ns and the TLB access time, T be 100ns. Find out-

a) Effective memory access with TLB

b) Effective memory access without TLB

V. Given memory partitions of 10K, 50K, 20K, 30K, and 60K(in order), how would each of the algorithm First-fit, Best-fit and Worst-fit place processes of 21K, 41K, 11K

and 42K (in order)? Which algorithm makes the best use of memory?

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. Explain Linked and Indexed allocation method of disk. Also discuss their merits and demerits. Suppose that the disk rotates at 3600 rpm.

a) What is the average rotational latency of this disk drive?

b) What seek distance can be covered in the time that you found for part (a)?

II. Suppose that a disk drive has 500 cylinders, numbered from 0 to 499. The drive is currently serving a request at cylinder 154, and the previous request was at cylinder 102. The queue of pending requests in FIFO order is 80, 147, 291, 177, 94, 409, 102, 350, and 130 starting from the current head position. What is the total head distance (in cylinders) that the disk arm moves for the following disk scheduling algorithms:

a. SSTF

b. LOOK

III. What do you understand by Belady's Anomaly? Which popular page replacement algorithm suffers from the Belady's Anomaly? Consider the following page reference string-

1, 2, 3, 1, 4, 5, 6, 2, 1, 3, 2, 7, 6, 3, 4, 1, 2, 6

How many page faults would occur for the following replacement algorithms assuming 3 frames?

a) LRU

b) Optimal

IV. Explain the difference between Internal & External fragmentation? In a paged segmented system, a virtual address consists of 32 bits, of which 12 bits are used for displacement, 11 bits are a segment number and 9 bits are a page number. Calculate-

a) Page Size

b) Maximum No. of Pages

c) Maximum Segmented Size

d) Maximum No. of Segments